

A More Credible Approach to Parallel Trends

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This paper proposes tools for robust inference in difference-in-differences and event-study designs where the parallel trends assumption may be violated. Instead of requiring that parallel trends holds exactly, we impose restrictions on how different the post-treatment violations of parallel trends can be from the pre-treatment differences in trends (“pre-trends”). The causal parameter of interest is partially identified under these restrictions. We introduce two approaches that guarantee uniformly valid inference under the imposed restrictions, and we derive novel results showing that they have desirable power properties in our context. We illustrate how economic knowledge can inform the restrictions on the possible violations of parallel trends in two economic applications. We also highlight how our approach can be used to conduct sensitivity analyses showing what causal conclusions can be drawn under various restrictions on the possible violations of the parallel trends assumption.

Key words: Difference-in-differences, Event-study, Parallel trends, Robust inference, Sensitivity analysis, Partial identification

JEL codes: C1

1. INTRODUCTION

Researchers using difference-in-differences (DiD) and related methods are often unsure about the validity of the parallel trends assumption needed for point identification of the causal parameter of interest. It has therefore become common practice to assess the plausibility of the parallel trends assumption by testing for pre-treatment differences in trends (“pre-trends”). Although pre-trends tests are intuitive, recent research has shown that they may suffer from low power (Freyaldenhoven *et al.*, 2019; Bilinski and Hatfield, 2020; Kahn-Lang and Lang, 2020; Roth, 2022), and that conditioning the analysis on passing pre-trends tests introduces statistical issues related to pre-testing (Roth, 2022). How then should researchers proceed when they are unsure about the validity of the parallel trends assumption?

This paper proposes methods for robust inference and sensitivity analysis in empirical settings where the parallel trends assumption may not hold. Building on work by Manski and Pepper (2018), we show that the causal parameter of interest can be (partially) identified under

a large class of restrictions that impose that the post-treatment violations of parallel trends cannot be “too different” from the pre-trends. We then introduce methods that yield uniformly valid inference for the treatment effect under the imposed restrictions. Intuitively, our inference methods account for both statistical uncertainty (we can only noisily estimate the true pre-trend) as well as “identification uncertainty” (even if the true pre-trend were known, we may not know exactly how to extrapolate it). Our approach thus formalizes the intuition motivating pre-trends testing while avoiding the statistical issues described above.

More concretely, we consider a setting in which the researcher estimates a vector of “event-study” coefficients $\hat{\beta} = (\hat{\beta}'_{pre}, \hat{\beta}'_{post})' \in \mathbb{R}^{T+\bar{T}}$, where $\hat{\beta}_{pre}$ and $\hat{\beta}_{post}$ respectively correspond with estimates for $\bar{T}$ pre-treatment periods and $\bar{T}$ post-treatment periods. We assume that $\hat{\beta}$ is consistent for the reduced-form parameter β , which can be decomposed as

$$\beta = \underbrace{\begin{pmatrix} 0 \\ \tau_{post} \end{pmatrix}}_{=: \tau} + \underbrace{\begin{pmatrix} \delta_{pre} \\ \delta_{post} \end{pmatrix}}_{=: \delta}, \quad (1)$$

where τ is a causal parameter of interest that is assumed to be 0 in the pre-treatment period and δ is a bias from a difference in trends. For instance, in the canonical (non-staggered) DiD framework, $\hat{\beta}$ may be the coefficients from an “event-study regression” specification, τ the vector of period-specific average treatment effects on the treated (ATT) for some policy of interest, and δ the difference in trends of untreated potential outcomes between the treated and comparison groups. As we discuss in Section 2, this framework also applies to more complicated empirical settings, such as those with staggered treatment timing (e.g. Callaway and Sant’Anna, 2020; Sun and Abraham, 2020). The usual parallel trends assumption used to point identify τ_{post} is that $\delta_{post} = 0$, and researchers frequently assess the plausibility of this assumption by testing the null hypothesis $\delta_{pre} = 0$ (a “pre-trends” test).

Instead of imposing that the parallel trends assumption holds exactly, we place restrictions on the possible post-treatment differences in trends δ_{post} given the point identified pre-trends δ_{pre} . Such restrictions formalize the intuition motivating pre-trends tests, namely that pre-trends are informative about counterfactual post-treatment differences in trends. Formally, we assume that $\delta \in \Delta$ for some researcher-specified set Δ , and show that the causal parameter τ_{post} is partially identified under such restrictions.

Restrictions of this form can be used to formalize a wide variety of intuitions about possible violations of the parallel trends assumption that are commonly expressed in applied work. For example, as discussed in Manski and Pepper (2018), researchers may be willing to assume that the confounding factors that create post-treatment violations of parallel trends are similar in magnitude to those in the pre-treatment period. This intuition can be formalized by specifying a Δ that bounds the maximal post-treatment violation of parallel trends by a parameter $\bar{M}$ times the maximal pre-treatment violation of parallel trends. In other contexts, researchers are concerned about violations of parallel trends from secular trends that are assumed to evolve smoothly over time. This intuition can be formalized by bounding the extent to which the slope of the violation of parallel trends can change over time. We adopt a flexible framework that allows researchers to capture these intuitions, as well as many other restrictions that are implied by context-specific knowledge about possible confounding factors.

We then introduce methods to conduct uniformly valid inference on a scalar parameter of the form $\theta = l'\tau_{post}$ under the restriction $\delta \in \Delta$. As emphasized in the recent literature on pre-trends testing, the pre-treatment coefficients $\hat{\beta}_{pre}$ are often imprecise estimates of δ_{pre} . It is therefore important to introduce inference methods that account for the statistical uncertainty

in the estimation of the event-study coefficients. We introduce two main inference approaches, with different desirable properties depending on the shape of Δ .

We first introduce a general inference approach that can accommodate a large class of choices for Δ . This approach is based on the observation that conducting inference on θ can be cast as the problem of testing a system of moment inequalities, allowing us to leverage a large econometrics literature on moment inequality testing (Canay and Shaikh, 2017 provide a recent review). The moments have a potentially large number of nuisance parameters that enter linearly, and we therefore consider an implementation of this approach based on the conditional and hybrid approaches of Andrews *et al.* (forthcoming, henceforth ARP), who considered moment inequalities with this structure. Uniform size control for these tests follows nearly immediately from results in ARP.

We then prove that the tests proposed by ARP have some desirable power properties in our context. First, the conditional and hybrid tests are consistent, in the sense that they have power approaching 1 against fixed alternatives outside of the identified set. Second, we prove that the conditional test has optimal local asymptotic power under a linear independence constraint qualification (LICQ) assumption. As described in Kaido *et al.* (2021), LICQ and related constraint qualifications have been used widely in the partial identification literature, and are often imposed to ensure size control. By contrast, we show that the ARP conditional test is asymptotically valid even when LICQ fails, but has optimal local asymptotic power when LICQ is satisfied. Intuitively, this result implies that the conditional test will perform well when the binding and non-binding moments are “far apart” relative to the sampling variation in the data. We provide several intuitive examples to illustrate when this result will and will not be applicable. Our result also implies that the ARP hybrid test will have near-optimal local asymptotic power under LICQ. These power results are new, and exploit additional structure in our context not contained in ARP.

Our second approach to inference is based on fixed length confidence intervals (FLCIs) (Donoho, 1994). FLCIs have desirable finite-sample guarantees for particular Δ s of interest. In particular, results from Armstrong and Kolesár (2018, 2020b) imply that when Δ is convex and centrosymmetric, FLCIs have near-optimal expected length in the finite-sample normal model. These results are applicable for one of our leading examples, Δ^{SD} , which restricts the smoothness of the difference in trends. In Monte Carlo simulations, we find that the use of such FLCIs can lead to substantial power gains over the conditional/hybrid approaches for Δ^{SD} when the length of the identified set is short relative to the sampling variation in the data. This is intuitive since the asymptotic power guarantees for the conditional/hybrid approaches are in the asymptotic regime where sampling uncertainty is small relative to the length of the identified set, in contrast to the finite-sample guarantees for FLCIs. On the other hand, FLCIs are applicable for a much smaller class of Δ s: indeed, we show that for many other choices of Δ , they will be inconsistent in the strong sense that power against fixed points outside the identified set need not converge to one asymptotically.

Based on our theoretical results and Monte Carlo simulations, we recommend the ARP hybrid approach for general forms of Δ , but prefer the FLCI approach in special cases (such as for Δ^{SD}) where the conditions for consistency and finite-sample near-optimality are met.

We recommend that applied researchers use our methods to construct robust confidence sets under economically-motivated restrictions on how the pre-trends relate to the post-treatment violations of parallel trends. Our tools can also be used to conduct sensitivity analyses in which the researcher reports confidence sets under varying restrictions on the possible differences in trends. For example, if the researcher suspects that the confounding factors in the post-treatment periods are similar in magnitude to those in the pre-treatment periods, then it may be reasonable to impose that the post-treatment violations of parallel trends are no larger than the maximum

pre-treatment violation of parallel trends. As a sensitivity analysis, the researcher might also report confidence sets that allow the maximum post-treatment violation of parallel trends to be up to $\bar{M}$ times larger than the maximum pre-treatment violation for different values of $\bar{M}$. The proposed sensitivity analysis thus follows the “top-down” approach described by Tamer (2010), in which the researcher reports what can be learned under a sequence of progressively weaker assumptions. Performing such sensitivity analyses makes clear what must be assumed about the possible differences in trends in order to draw specific causal conclusions, in line with the partial identification paradigm initiated by Charles Manski and co-authors in a variety of contexts (Manski, 1989, 1990, 2003, 2008). We provide the HonestDiD R and Stata packages to implement our methods.¹ We illustrate our recommended approach with applications to two recently published papers, in which we show how the choice of the restrictions Δ can be tailored to the economic context.

1.1. Related literature

The approach in this paper builds on the foundational partial identification analysis for DiD in Manski and Pepper (2018). Manski and Pepper consider identification under researcher-specified bounds on the magnitude of δ_{post} (what they call “bounded DiD variation”), and calibrate these bounds using the maximal pre-treatment violation of parallel trends in their empirical application on the effects of right-to-carry gun laws.² One of our leading classes of restrictions, Δ^{RM} , formalizes this calibration approach by bounding the magnitude of post-treatment violations of parallel trends by $\bar{M}$ times the maximal pre-treatment violation. Our framework also allows for many other intuitive restrictions—such as bounds on how far δ can deviate from linearity—and it can be applied to a variety of DiD estimators, including recent proposals for settings with staggered treatment timing. Most importantly, while Manski and Pepper (2018) provide a framework for identification, we provide inference methods to construct uniformly valid confidence sets for the treatment effect of interest. This allows applied researchers to account for statistical uncertainty in their analyses, which can be important since event-study coefficients are often imprecisely estimated in practice.

Several other recent papers consider various relaxations of the parallel trends assumption. Keele *et al.* (2019) develop techniques for testing the sensitivity of DiD designs to violations of the parallel trends assumption, but they do not incorporate information from the observed pre-trends in their sensitivity analysis. Empirical researchers commonly adjust for the extrapolation of a linear trend from the pre-treatment periods when there are concerns about violations of the parallel trends assumption, which is valid if the difference in trends is exactly linear (*e.g.* Bhuller *et al.*, 2013; Dobkin *et al.*, 2018; Goodman-Bacon, 2018, 2021). Our methods nest this approach as a special case, but allow for valid inference under less restrictive assumptions about the class of possible differences in trends (such as when δ is only approximately linear). Freyaldenhoven *et al.* (2019) propose a method that allows for violations of the parallel trends assumption but requires an additional covariate that is affected by the same confounding factors as the outcome but not by the treatment of interest. Ye *et al.* (2020) consider partial identification of treatment effects when there exist two control groups whose outcomes have a bracketing relationship with

1. The latest version of the R and Stata packages are respectively available at <https://github.com/asheshrambachan/HonestDiD> and <https://github.com/mcaceresb/stata-honestdid/>.

2. Manski and Pepper (2018) also consider “bounded time” and “bounded state” restrictions that bound how much the mean of $Y(0)$ can differ either across treatment groups or within-groups over time. Such restrictions could also be incorporated into our framework by augmenting the vector $\hat{\beta}$ to include group-specific sample averages.